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github.com/the-deep-33/crypto-direction-nn

Undergraduate Thesis Proposal

Directional Prediction of Daily BTC/USDT Returns under Transaction-Cost-Adjusted Evaluation

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Topic and Framing

Directional prediction of daily BTC/USDT returns using machine learning, evaluated under walk-forward validation and against a transaction-cost-adjusted decision threshold.

One change since our last conversation

The project now includes a volatility component. The reason is the standard decomposition of a return into scale and shock:

$$r_t = \sigma_t z_t$$

The scale σ_t is strongly predictable — volatility clustering is among the most robust regularities in financial data. The sign is not: since $\sigma_t > 0$, we have $\text{sign}(r_t) = \text{sign}(z_t)$ exactly, and a variance model therefore contributes nothing to direction.

The two halves consequently do not compete. The volatility model produces the conditional forecast $\hat{\sigma}_t$, which performs three distinct roles in the directional analysis:

- **Feature standardisation.** A 3% move means something different in a calm regime than in a turbulent one. Dividing return-scaled features by $\hat{\sigma}_t$ makes them regime-invariant, which matters on a sample of roughly 1,400 daily observations.
- **Decision threshold.** It sets the probability the classifier must exceed for a trade to be worth taking.
- **Position sizing and evaluation.** It provides volatility-targeted sizing and a correctly time-varying risk denominator for performance metrics.

Success Criteria

The project brief gives 51–55% accuracy as marginally profitable, 55–60% as profitable, and above 60% as practically applicable. I would like to make the economics behind these bands explicit rather than assumed.

Let $\hat{m}_t$ denote the expected absolute move and c the one-way transaction cost, both as fractions. A directional position is correct with probability p and captures approximately $\hat{m}_t$; it is wrong with probability $1 - p$ and loses approximately $\hat{m}_t$. The round trip costs $2c$. Expected profit is therefore

$$\mathbb{E}[\text{profit}] = \hat{m}_t (2p - 1) - 2c$$

Setting this to zero gives the break-even probability

$$p_t^* = \frac{1}{2} + \frac{c}{\hat{m}_t}$$

This threshold is not a constant. Taking $c = 0.10\%$ one-way and $\hat{m}_t \approx 0.8 \hat{\sigma}_t$:

Regime	$\hat{\sigma}_t$	$\hat{m}_t$	p_t^*
Calm	1.5%	1.20%	0.583
Normal	2.5%	2.00%	0.550
Turbulent	4.0%	3.20%	0.531

The same accuracy is therefore worthless in one regime and profitable in another. The bands hold for a given volatility level rather than unconditionally. Deriving this threshold explicitly, and reporting accuracy conditional on regime rather than averaged across regimes, is what I would like the thesis to contribute.

On the expected result

I want to state this in advance rather than after the fact. Under honest walk-forward evaluation I expect unconditional accuracy in the low fifties, with a confidence interval that may include 50%. I am treating the accuracy figure as a measurement rather than a target.

The thesis succeeds if the evaluation is correct and the threshold analysis answers the conditional question: whether an ex-ante identifiable subset of days exists on which model confidence clears p_t^* , and whether that subset is large enough to be economically meaningful. A negative answer, properly measured and properly bounded, is a result I am prepared to report and defend.

Scope

To meet the September deadline, the following are deliberately excluded from the thesis and held for subsequent work: pooling across multiple assets, GJR-GARCH and the volatility asymmetry analysis, sequence models, sentiment features, and formal data-snooping tests (block bootstrap, White's Reality Check / Hansen's SPA, deflated Sharpe ratio).

These are the natural continuation and the basis on which the work could be developed into a paper after the defence.

Proposed Structure

1. **Introduction** — problem statement and motivation; why direction rather than price level; research question and hypothesis; contribution and structure.
2. **Literature Review** — market efficiency and the predictability debate; technical indicators and price patterns; machine learning in financial forecasting; volatility modelling from ARCH to GARCH; methodological pitfalls (leakage, look-ahead bias, data snooping).
3. **Data** — source, instrument, period and frequency; integrity checks (monotonicity, duplicates, gaps); target definition as the sign of the next-day log return; class balance and its implications for the baseline.
4. **Methodology** — walk-forward validation and why standard k -fold is invalid on time series; the decomposition $r_t = \sigma_t z_t$; conditional volatility from ARCH to GARCH(1,1) and the persistence parameter; derivation of the time-varying threshold p_t^* ; evaluation metrics (accuracy, AUC, calibration, cost-adjusted return); feature engineering and the case for volatility standardisation.
5. **Exploratory Data Analysis** — return distribution, fat tails and skew; autocorrelation of r_t versus autocorrelation of r_t^2 ; visual evidence of volatility clustering; regime structure over the sample period.
6. **Volatility Modelling** — realised volatility estimators (Parkinson, Garman-Klass); GARCH(1,1) fitted inside each walk-forward window; one-step-ahead forecasts and the in-window fitting protocol; diagnostics on standardised residuals and Ljung-Box on $\hat{z}_t^2$.
7. **Directional Modelling** — baselines (majority class, momentum rules); logistic regression; gradient boosting; neural network; identical harness and identical features, raw versus volatility-standardised.
8. **Results** — volatility model performance and diagnostics; unconditional directional accuracy with confidence intervals; accuracy conditional on volatility regime; the p_t^* series, coverage curve and hit rate on qualifying days; trading simulation with costs and volatility-targeted sizing.
9. **Discussion** — interpretation of the conditional result; limitations (sample size, single asset, absence of formal snooping tests); what the sample does and does not support.
10. **Conclusion and Further Work**

#	Experiment and purpose	Outcome known in advance?
E1	Baselines under walk-forward. Majority class, momentum rules, logistic regression. Establishes the floor any subsequent model must beat.	Largely — expect 50–52%
E2	Realised volatility estimators. Parkinson and Garman-Klass compared against squared returns. Establishes a lower-noise volatility proxy from OHLC data.	Yes — both are known to be more efficient
E3	GARCH(1,1) in-window. Fitted inside each walk-forward window; one-step-ahead forecasts only; Ljung-Box test on $\hat{z}_t^2$. Produces $\hat{\sigma}_t$ and verifies the variance dynamics are captured.	Largely — GARCH is expected to fit well
E4	Accuracy by volatility regime. Directional accuracy computed within volatility quantile buckets. <i>The central test: does predictive edge concentrate in a particular regime?</i>	No — genuinely open
E5	Model ladder. Logistic regression, gradient boosting and neural network under an identical harness, on raw versus volatility-standardised features. Determines whether flexibility helps and whether standardisation helps.	Partly — expect a flat or near-flat ladder
E6	Threshold and coverage analysis. The p_t^* series, the coverage curve, and the hit rate on qualifying days. Converts accuracy into an economic decision.	No
E7	Trading simulation. Transaction costs and volatility-targeted position sizing applied to the signals from E4 and E6. Establishes economic significance.	No

Note on E4

This is the experiment whose outcome I do not know. If accuracy varies materially across volatility regimes, the conditional story has substance and E6 and E7 follow naturally from it. If accuracy is flat across regimes, I will report that as a null result: the threshold machinery still applies, but the claim becomes correspondingly narrower. I would rather state this before running the experiment than after.

Note on E5

The neural network is included as the flexible end of a comparison ladder rather than as the centrepiece of the thesis. The informative outcome is the *shape* of the ladder. If added flexibility does not improve performance, that is evidence about the structure of the signal rather than a failure of the model.

Open Questions

- 1. Administrative requirements.** What is required procedurally — topic registration, committee formation, any deadlines at the student office — and by when? This appears more likely to constrain a defence before 28 September than the writing itself.
- 2. Confirmation of direction.** I am beginning to write on this structure now and will proceed unless you indicate otherwise. Could we arrange twenty minutes online before 11 September to confirm?